

# LAB 4 – VIBECODING

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## INITIAL PROMPT – PROJECT SETUP

I am completing Lab 4 – VibeCoding using a Python-based Rock Paper Scissors game. I need to work through the repository README, fix the existing broken functionality, and implement the listed features one task at a time. Please make the required changes without unnecessarily modifying unrelated files or existing game functionality.

## TASK 1 – FIX WINNER EVALUATION

Fix the broken winner evaluation in the Rock Paper Scissors game. The current `determine_winner()` logic has the winning and losing results inverted. Correct the mappings so that Rock beats Scissors, Scissors beats Paper, and Paper beats Rock, while identical choices remain a tie. Keep the existing scoring and user interface unchanged.

## TASK 2 – FIRST-TO-X MATCH VICTORY

Implement the first-to-X match feature for the Rock Paper Scissors game, using 3 as the target score. The match should end when either the player or CPU reaches 3 points and display a clear GAME OVER message showing the winner. Prevent additional moves after the match ends, and allow the player to press R to reset the scores and start a new match.

## TASK 3 – ADAPTIVE CPU

Add an adaptive CPU strategy to the Rock Paper Scissors game. Track the player's move history during the current match, identify the move selected most frequently, and make the CPU favor the move that counters that pattern: Paper against Rock, Scissors against Paper, and Rock against Scissors. Keep some randomness so the CPU does not counter every time, and reset the move history when a new match starts.

## TASK 4 – PROCEDURAL ICONS AND SHAKE ANIMATION

Add procedural graphical icons for Rock, Paper, and Scissors using Pygame drawing primitives instead of external image files. When a player makes a move, show a short shake animation before revealing the player's and CPU's selected moves. Keep the existing scoring, winner evaluation, first-to-3 match system, GAME OVER state, reset functionality, and adaptive CPU behavior working correctly.